

2.3 COELOM

The term 'coelom' was coined by **Haeckel**. The body cavity, which is lined by mesoderm, is called coelom. More elaborately, coelom is a fluid-filled space between the body wall and visceral organs and lined by mesodermal epithelium, the peritoneum. Animals possessing coelom are called coelomates/eucoelomates. Evolution of efficient organ systems was not possible until the evolution of coelom for supporting the organs and distributing material.

2.3.1 Acoelomate bilaterians

The bilaterian animals in which the body cavity is absent are called acoelomates, e.g. Platyhelminthes (lowest bilaterians). In these animals, the mesenchyme derived from the third germinal layer, called mesoderm, occupies the entire blastocoel,

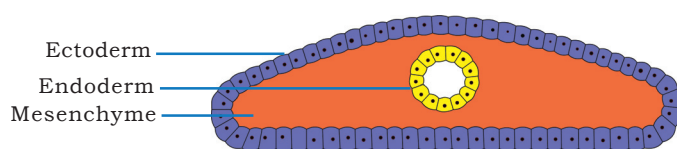


Fig. 2.5 T.S of acoelomate

between the ectoderm and the endoderm, so that the adults have neither the primary cavity (blastocoelom) nor the secondary cavity (coelom). As there is no body cavity, the acoelomates exhibit **solid body plan**. Problems faced by the

acoelomates due to absence of perivisceral cavity are - their internal organs cannot move freely, as they are embedded in the mesenchyme, diffusion of material from the gut to the body wall is slow and less efficient.

2.3.2 Pseudocoelomate bilaterians

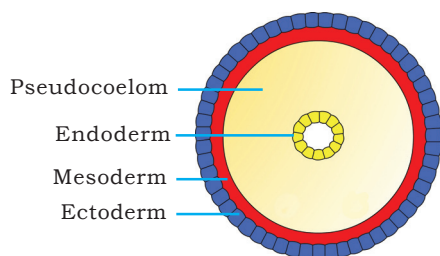


Fig. 2.6 T.S of pseudocoelomate

In some animals, the body cavity is not lined by mesodermal epithelia. Such animals are called Pseudocoelomates. They include the members of phylum Aschelminthes (Nematoda, Rotifera and some minor phyla). During the embryonic development, mesoderm (mesenchyme) occupies only a part of the blastocoel adjoining the ectoderm. The unoccupied portion of the blastocoel persists as **pseudocoelom**, which is filled with

pseudocoelomic fluid. Pseudocoelomates are the first animals to exhibit a '**tube-within-a-tube**' organisation.

Cnidarians and some flatworms have an incomplete gut and exhibit 'blind sac plan'. Justify the statement. Animals from nematodes to chordates in general have a complete gut and exhibit 'tube - within-a-tube plan'. Explain.

As the gut wall is made of only endodermal epithelium, diffusion of digested food from the lumen of the gut into the surrounding pseudocoelomic fluid becomes

easier and the absence of circulatory system is thus compensated. Though it is called pseudocoelom (false coelom), it performs almost all the functions of a regular coelom. Pseudocoelomic fluid of pseudocoelomates serves as a 'hydrostatic skeleton' and a 'shock absorber'. It allows the free movements of visceral organs, helps in the circulation of nutrients, and storage of nitrogenous wastes.

"Pseudocoelom is considered a false body cavity." Substantiate the statement?

2.3.3 Eucoelomate bilaterians

Coelom or 'true coelom' is a fluid-filled cavity, that lies between the body wall and the visceral organs and is lined by mesodermal epithelium, the peritoneum. The portion of the peritoneum that underlines the body wall is the **parietal peritoneum** or **somatic peritoneum**. The portion of the peritoneum that covers the visceral organs is the **splanchnic peritoneum** or **visceral peritoneum**. In coelomates, the visceral organs are suspended in the coelom by the peritoneum. A double layered peritoneum that connects some visceral organs to the body wall is called **mesentery**. In some eucoelomates such as the annelids, the dorsal and ventral mesenteries divide the coelom into paired compartments. Certain organs such as the kidneys of the vertebrates are covered by the parietal peritoneum only on their ventral side. Such a peritoneum is called the 'retroperitoneum' and the organs lined by it are called '**retroperitoneal organs**'.

During the embryonic development of the eucoelomates, the blastocoel is replaced by true coelom derived from the mesoderm. So, the true coelom is also called '**secondary body cavity**'. In the eucoelomates, mesoderm lines both the body wall and the walls of the visceral organs. So, visceral organs become 'muscular' and exhibit free movements 'independent' of the movement of the body wall in the coelomic fluid. As the wall of the gut becomes thick and muscular, digested food and other nutrients cannot easily diffuse from the lumen of the gut into the coelom i.e. fluid. Circulatory system (blood vascular system) is developed in the eucoelomates to overcome this problem. Based on the mode of formation of coelom, the eucoelomates are classified into two types:

I. Schizocoelomates

Animals in which the body cavity is formed by 'splitting of mesoderm' are called **schizocoelomates**. Annelids, arthropods and molluscs are schizocoelomates in the animal kingdom. All the schizocoelomates are **protostomians** and they show '**holoblastic**', '**spiral**' and '**determinate**' cleavage. **The 4d blastomere** or **mesentoblast cell** of the early embryo divides to form mesodermal blocks between the ectoderm and the endoderm and replaces the blastocoel. The split that appears in each

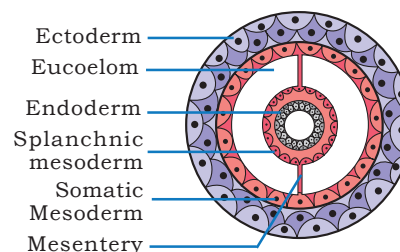


Fig. 2.7 T.S of eucoelomate

mesodermal block leads to the formation of '**schizocoelom**' (**split coelom**). In the annelids, the functional body cavity (perivisceral cavity) is schizocoelom. It is in the form of a series of paired coelomic cavities, but in arthropods and molluscs, the functional body cavity that lies around visceral organs is filled with blood (haemolymph) and is called **haemocoel**. It is formed by the fusion of the embryonic blastocoel with some true coelomic spaces and the tissues are directly bathed in the blood (haemolymph).

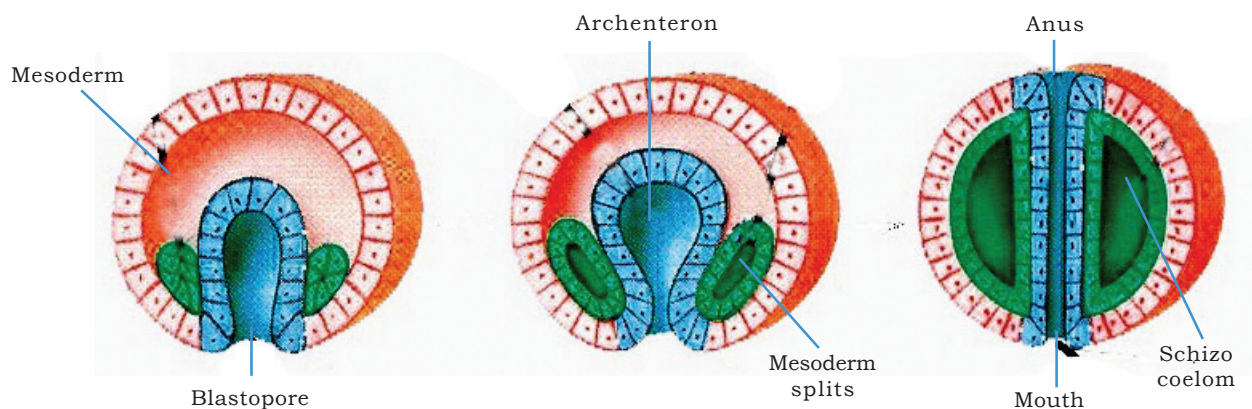


Fig. 2.8 Development of schizocoelom

II. Enterocoelomates

Animals in which the body cavity is formed from the mesodermal pouches of archenteron are called **enterocoelomates**. Echinoderms, hemichordates and chordates are the enterocoelomates. In these animals, mesodermal pouches that evaginate from the wall of the archenteron into the blastocoel are fused with one another to form the enterocoelom. All the enterocoelomates are **deuterostomes** and they show **radial** and **indeterminate** cleavage.

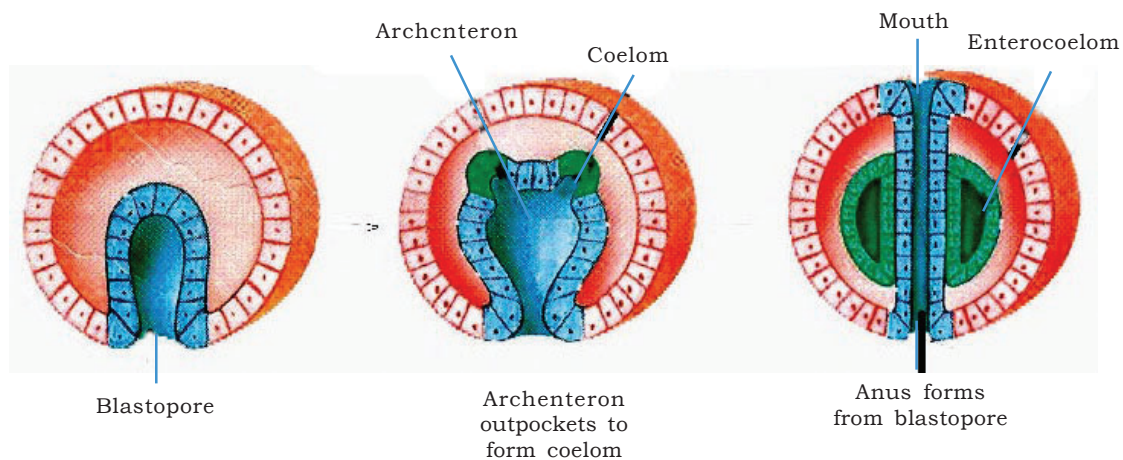
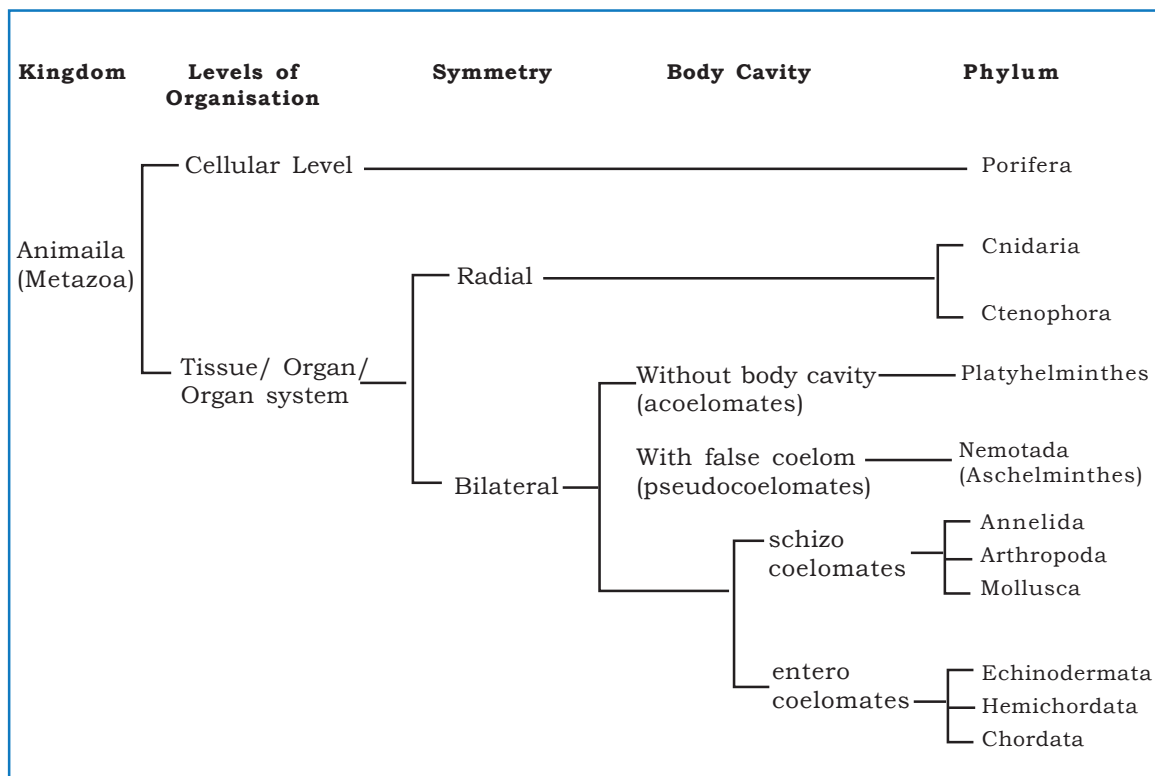


Fig. 2.9 Development of enterocoelom

Advantages of coelom over pseudocoelom

1. Visceral organs of eucoelomates are muscular (because of their association with mesoderm) and so they can contract and relax freely independent of the muscular movements of the body wall in the coelomic space, e.g. peristaltic movements of alimentary canal.
2. Gametes are released into the coelom in some invertebrates (which do not have gonoducts) and in the female vertebrates.
3. In non-chordates, coelomic fluid receives excretory products and stores them temporarily before their elimination.
4. In the eucoelomates, the mesoderm comes into contact with the endoderm of the alimentary canal, and it causes 'regional specialization of the gut, such as the development of gizzard, stomach etc. This is referred to as '**primary induction**'. In the case of the pseudocoelomates, due to the absence of such a contact between the gut and the mesoderm, the wall of the gut does not show complex and highly specialized organs.

The presence of a simple, thin walled gut in the nematodes allows easy diffusion of the digested foods into the pseudocoelomic fluid, thus compensating for the absence of circulatory system.



Summary of organisational levels, symmetry and body cavity